

Vehicle Sales Data Analysis

1. Project Overview:

This project analyses vehicle sales data to understand customer purchasing patterns, product line performance, regional sales trends, and revenue distribution. By examining historical sales transactions, the project aims to identify top customers, high-growth markets, best-selling vehicle categories, and the impact of deal sizes on revenue generation. The insights derived can help optimize inventory planning, strengthen customer retention strategies, and drive business growth.

2. Dataset Summary:

Rows: 2,833

Columns: 13

- **Order Information:** Order Number, Order Date, Order Line Number, Order Status
- **Sales Metrics:** Quantity Ordered, Price Each, Total Sales (Derived Metric)
- **Product Details:** Product Code, Product Line
- **Customer Information:** Customer Name, Contact Name
- **Geographic Data:** Country, State
- **Deal Information:** Deal Size (Small, Medium, Large)
- **Business Performance Indicators:** Revenue, Product Demand, Customer Contribution, Market Performance

3. Exploratory Data Analysis using Python:

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	ORDERDATE	\
	0	10107.0	30.0	95.70	2.0	2/24/2003 0:00
	1	10121.0	34.0	81.35	5.0	05-07-2003 00:00
	2	10134.0	41.0	94.74	2.0	07-01-2003 00:00
	3	10145.0	45.0	83.26	6.0	8/25/2003 0:00
	4	10159.0	49.0	100.00	14.0	10-10-2003 00:00

	STATUS	QTR_ID	MONTH_ID	YEAR_ID	PRODUCTLINE	...	\
	0	Shipped	1.0	2.0	2003.0	Motorcycles	...
	1	Shipped	2.0	5.0	2003.0	Motorcycles	...
	2	Shipped	3.0	7.0	2003.0	Motorcycles	...
	3	Shipped	3.0	8.0	2003.0	Motorcycles	...
	4	Shipped	4.0	10.0	2003.0	Motorcycles	...

	ADDRESSLINE1	ADDRESSLINE2	CITY	STATE	POSTALCODE	\
	0	897 Long Airport Avenue	NaN	NYC	NY	10022
	1	59 rue de l'Abbaye	NaN	Reims	NaN	51100
	2	27 rue du Colonel Pierre Avia	NaN	Paris	NaN	75508
	3	78934 Hillside Dr.	NaN	Pasadena	CA	90003
	4	7734 Strong St.	NaN	San Francisco	CA	NaN

	COUNTRY	TERRITORY	CONTACTLASTNAME	CONTACTFIRSTNAME	DEALSIZE
	0	USA	NaN	Yu	Kwai Small
	1	France	EMEA	Henriot	Paul Small

- **Missing Data Handling:** Checked for null values and imputed missing values in the Order Number column using the mean value of each.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:** Created contact name column by binning first name and last name.
- **Data Consistency Check:** Verified Orderdate.. dropped 'Qtr_id', 'Month_id', 'Year_id' as they were not needed.

4. Data Analysis using SQL (Business Transactions):

- **Revenue by Country** – Compared total revenue generated by top 5 countries.

country	Total_Sales
USA	2986425.21
Spain	1021705.97
France	919257.85
Australia	521598.46
UK	413203.34

- **High-Spending Users** – Identified customers who made the greatest number of purchases.

customername	Total_Sales
Euro Shopping Channel	766195.05
Mini Gifts Distributors Ltd.	530587.19
Australian Collectors, Co.	164575.87
La Rochelle Gifts	153909.82
Muscle Machine Inc	151782.04
Land of Toys Inc.	138798.17
AV Stores, Co.	136873.25
The Sharp Gifts Warehouse	136383.04
Souvenirs And Things Co.	133223.38
Dragon Souvenirs, Ltd.	131279.95

- **Top 5 Products** – Found products that were hot selling.

productline	Total_Sales
Classic Cars	2968546.4
Vintage Cars	1644212.05
Motorcycles	971086.29
Trucks and Buses	947355.18
Planes	877942.21

- **Best Product Line in Every Country:** Found the best-selling vehicle according the country

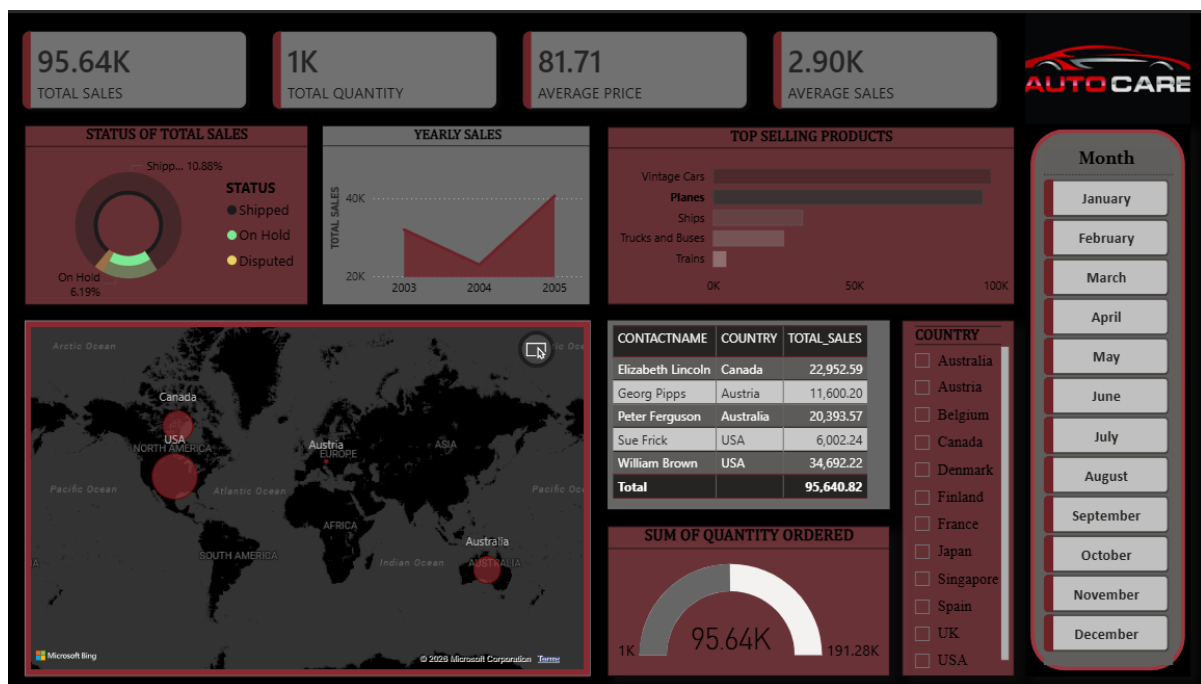
COUNTRY	PRODUCTLINE	revenue
USA	Classic Cars	1015261.93
Spain	Classic Cars	369361.76
France	Classic Cars	302423.44
Australia	Vintage Cars	157937.06
UK	Classic Cars	127062.04
Finland	Classic Cars	114522.78
Germany	Classic Cars	113357.71
Denmark	Classic Cars	113343.46
Norway	Classic Cars	101637.52

- **Revenue Lost Due to Cancelled / Non-Shipped Orders:** We found the revenue leakage caused due to cancellation or shipment issue.

STATUS	revenue
Cancelled	171195.49
Disputed	55171.32
In Process	120601.78
On Hold	158910.2
Resolved	133868.64

5. Dashboard in Power BI:

Finally, we have built an interactive Power BI Dashboard.



6. Business Growth Recommendations:

Sales Growth Recommendations-

- Focus marketing efforts on the top-performing vehicle categories to maximize revenue.
- Introduce promotional campaigns for low-selling vehicle models to boost demand.

Customer-Focused Recommendations-

- Develop loyalty programs to encourage repeat purchases.
- Analyse customer demographics and tailor marketing campaigns accordingly.

Revenue Optimization-

- Bundle vehicles with accessories, maintenance packages, or insurance offers.
- Review pricing strategies for low-margin products.